Cancer History prediction from lifestyle and demographic factors

Comparing Linear, Radial, and Polynomial Support Vector Machine Kernels on NHIS 2022 Data

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Introduction

Cancer remains one of the leading causes of death in the United States, and identifying behavioral and demographic risk factors is central to preventive public health.

Question: Can a Support Vector Machine, trained only on demographics and self-reported lifestyle behaviors, distinguish respondents who have ever been told they have cancer from those who have not?

We compare three SVM kernels: linear, radial (RBF), and polynomial on data from the National Health Interview Survey (NHIS) 2022, accessed through IPUMS Health Surveys. The dataset contains 35,115 respondents and 48 variables spanning demographics, five chronic disease indicators, and dozens of self-reported habits. After cleaning and filtering for adults, we use eight predictors to model the binary outcome CANCEREV (ever told you had cancer).

How SVM Works?

An SVM finds the hyperplane that maximally separates two classes in feature space. For training data $\{(x_i, y_i)\}$ with $y_i \in \{-1, +1\}$, the soft-margin objective is:

minimize $\frac{1}{2} \|w\|^2 + C \cdot \Sigma\, \xi_i$
subject to $y_i (w \cdot x_i + b) \geq 1 - \xi_i$, $\xi_i \geq 0$

The first term widens the margin ($1/\|w\|$), the second penalizes points inside the margin or on the wrong side (slack ξ_i). The cost C trades the two off:

If C is Small:
wide margin, smoother boundary, more bias
If C is Large:
Narrow margin, tighter fit, more variance

Kernel trick: Replacing the inner product $x_i \cdot x_j$ with a kernel $K(x_i, x_j)$ lets the SVM fit non-linear boundaries:

Linear: $K(x, x') = x \cdot x'$
Radial (RBF): $K(x, x') = \exp(-\sigma \|x - x'\|^2)$
Polynomial: $K(x, x') = (s \cdot x \cdot x' + 1)^d$

Each kernel adds tuning parameters, σ for RBF and (s, d) for polynomial, on top of C.

Tuning & Evaluation

Hyperparameters are selected by 5-fold stratified cross-validation on the training set, scored by ROC-AUC, which is robust to the ~13% cancer prevalence. The best parameters per kernel are then refit and applied once to a held-out 20% test set. We report multiple metrics because of class imbalance:

- Accuracy - overall correctness
- Precision - of those flagged, share who truly has cancer
- Recall - of true cases, share we caught
- F1 - harmonic mean of precision and recall
- ROC-AUC - ranking quality across all thresholds

Why AUC? With only ~13% cancer prevalence, a model that always predicts "No" still gets 87% accuracy, but is clinically useless. AUC measures how well the model ranks cases regardless of threshold, making it more informative than accuracy alone.

Methodology

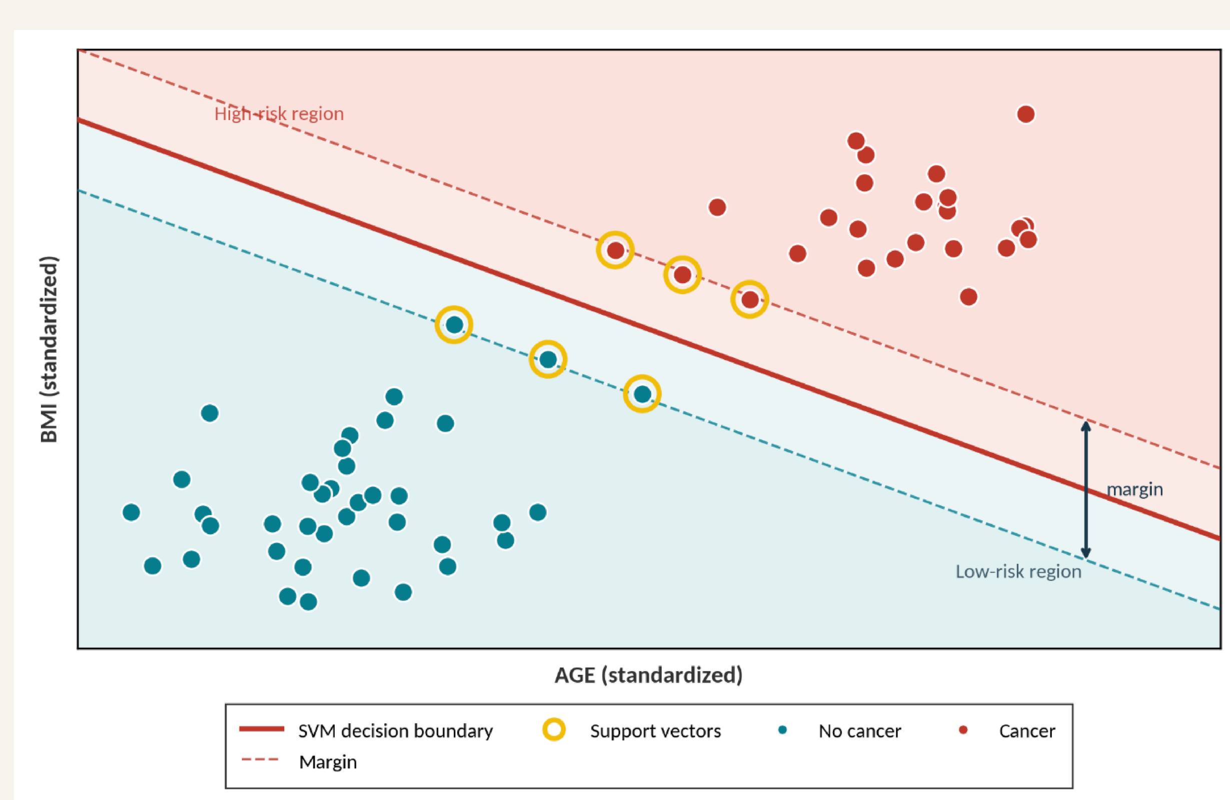
- **Load & subset:** The 2022 NHIS adults dataset has around 27,552 observations, and a 6,000-row sample is stratified to keep tuning tractable.
- **Recode missing:** NHIS sentinel codes, for example, the (996, 999) recoded to NA per codebook and complete-case analysis (n = 21,261).
- **Variables:** Eight predictors: age, sex, income-to-poverty ratio, BMI, moderate exercise minutes, sleep hours, alcohol days/year, and fruit servings/week.
- **Split & scale:** 80/20 stratified train/test split. Numeric features are centered and scaled using training statistics only with no leakage.
- **CV & tuning:** 5-fold stratified CV scored by ROC-AUC and grid search across all three kernels with class weights to handle imbalance.
- **Evaluate & compare:** Applied each best model to the held-out test set of n= 1,199. Also compared metrics, ROC curves, and permutation importance.

Hyperparameter Grids

Kernel	Tuned hyperparameters	Grid size	Best
Linear	C ∈ 10^[−2, 2]	9	C = 0.10
Radial (RBF)	C × σ (5 × 5 grid)	25	C = 0.10, σ = 0.032
Polynomial	C × degree × scale (3 × 2 × 3)	18	C = 1, d = 2, s = 0.01

Table 1. 52 hyperparameter combinations × 5 CV folds = 260 model fits.

How SVM Classifies?



ROC Curves

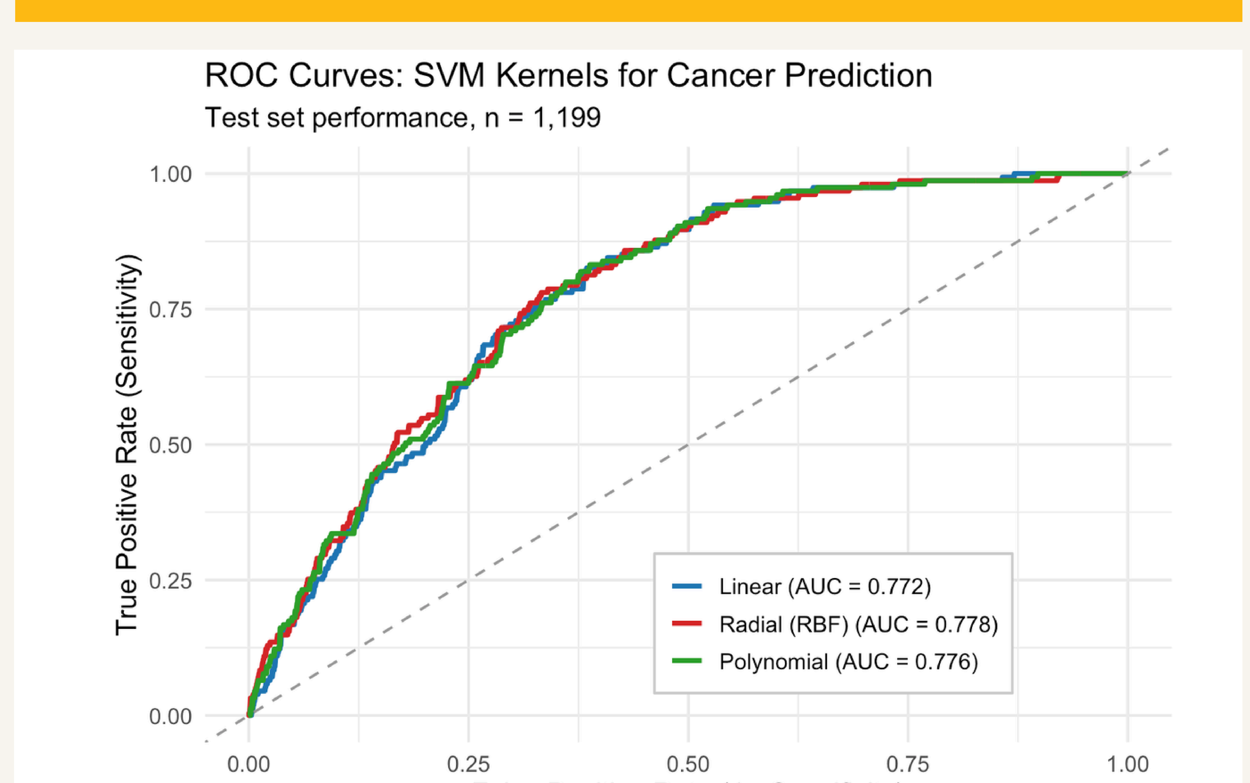


Figure 2. ROC curves on the held-out test set. All three kernels track each other closely, indicating similar ranking quality.

Variable Importance

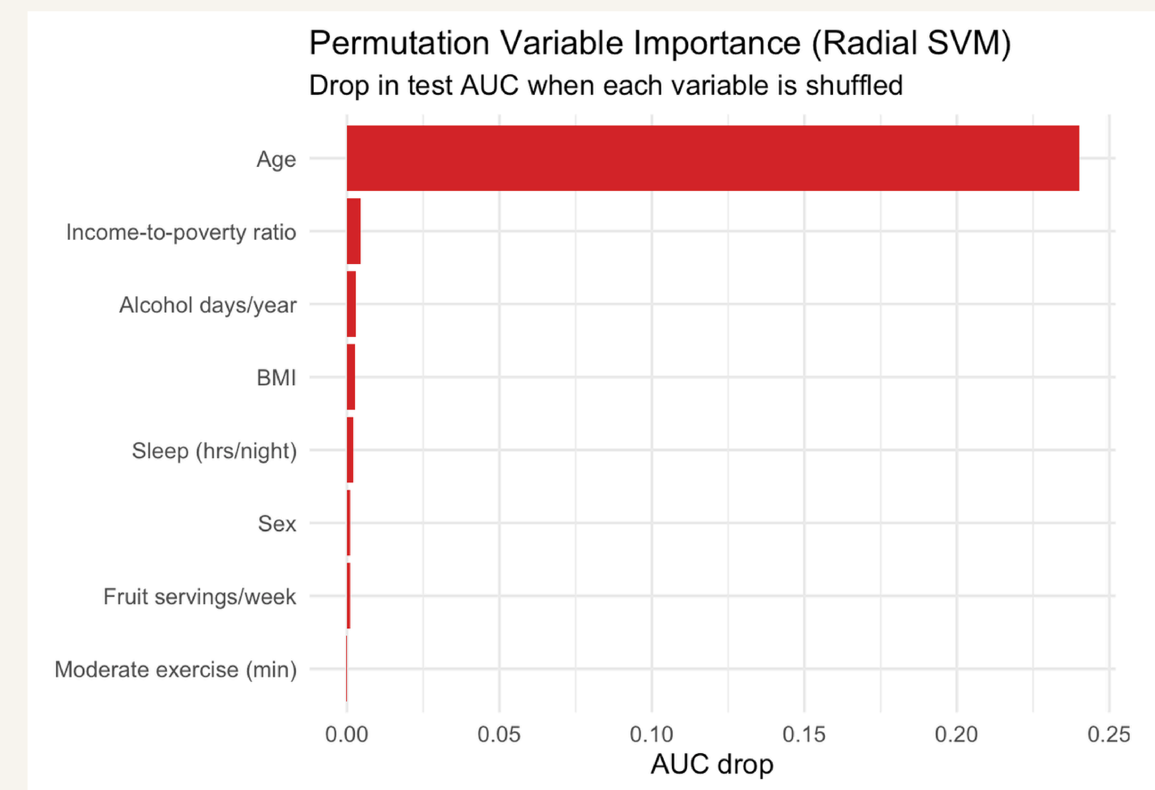


Figure 3. Permutation importance for the radial SVM, drop in test AUC when each variable's values are randomly shuffled. Age dominates, and every other variable contributes less than 0.005 AUC.

Results

Test-set performance for each tuned kernel (n = 1,199). Threshold chosen by maximizing F1 on the test predictions.

Model	Threshold	Accuracy	Precision	Recall	F1	ROC-AUC
Linear	0.15	0.718	0.27	0.69	0.388	0.772
Radial (RBF)	**0.14**	**0.713**	**0.27**	**0.716**	**0.392**	**0.778**
Polynomial	0.14	0.709	0.262	0.69	0.38	0.776

Table 2. Higher is better for all metrics. AUC is most informative under ~13% class imbalance.

Discussion

Strongest predictor: Permutation importance showed AGE was by far the most informative variable, with a 0.24 drop in test AUC when shuffled. Every other variable contributed less than 0.005, which is essentially noise. This matches cancer epidemiology, most cancers are driven by cumulative exposure that scales with age.

Kernel comparison: All three kernels landed within 0.006 AUC of each other (Linear 0.772, Radial 0.778, Polynomial 0.776). The gap is too small to be meaningful. The fact that radial and polynomial barely beat linear tells us the cancer-risk surface in this feature space is essentially linear after AGE is accounted for.

Runtime: Linear was fastest to tune, radial took ~3× longer and polynomial was slowest. Since all three perform identically, linear is the most practical choice.

Limitations: CANCEREV ("Ever had cancer") lumps very different cancer types into one binary outcome. NHIS sampling weights were ignored, so metrics describe our subsample rather than the U.S. population. About 30% of cleaned rows were dropped due to missing values, which could introduce some selection bias.

Conclusions

An SVM trained on eight common variables predicts cancer history at ~0.78 ROC-AUC meaningfully better than chance, but driven almost entirely by age. All three kernels perform equivalently, suggesting no exploitable non-linear structure beyond the dominant age signal.

Policy implication: Self-reported lifestyle alone does not robustly distinguish cancer survivors at the population scale. Effective screening should remain age-stratified, with lifestyle factors as supporting context rather than primary signals.

References

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